

Dr. rer. nat. Basavaraj Hampiholi

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[LinkedIn](#) | [GitHub](#)

PROFESSIONAL EXPERIENCE

Ulm University

03/2023 – Present

Scientific Research Associate | Medical AI & Data Science

- Designed and implemented an end-to-end, FAIR-aligned clinical data lake on Kubernetes, integrating MinIO object storage, scalable ETL pipelines, analytics tools, and ML frameworks to support reproducible clinical data analysis.
- PrivateAIM: Applied a federated learning framework for privacy-preserving analysis of distributed clinical data across multiple university hospitals, and executed an end-to-end pancreatic cancer use case to validate feasibility and analytical performance.
- Collaborated across technical and domain stakeholders to translate research requirements into robust data and ML infrastructure for clinical research.

BMW Car IT GmbH | Ulm, Germany

12/2018 – 02/2022

Ph.D. Student | Deep Learning and Computer Vision

- Developed and trained deep neural architectures for fine-grained human action segmentation in video, achieving state-of-the-art performance with improved computational efficiency.
- Proposed a transformer-based multimodal fusion approach for combining heterogeneous and homogeneous sensory modalities, improving representation learning for activity understanding tasks.
- Conducted research on multimodal learning, language-guided video understanding, and efficient deep learning architectures, with results published in peer-reviewed venues.

NEC Laboratories Europe | Heidelberg, Germany

04/2018 – 09/2018

Machine Learning Intern | Image Processing and Computer Vision

- Developed deep learning methods for automatic detection and tracking of mitotic cell-division phases in microscopy images, supporting biomedical image analysis for cancer research.
- Built video analytics models for urban monitoring applications, including automatic detection of illegal dumping in public environments.

Tata Consultancy Services (TCS) | Pune, India

12/2011 – 11/2014

Data Engineer | Manufacturing Information Systems and Data Analytics

- Built data ingestion and preprocessing pipelines using sensor, historian, planning, and quality-system data for industrial analytics use cases.
- Developed SQL- and R-based automation workflows for data extraction, cleaning, and reporting in production environments.
- Supported predictive analytics initiatives for manufacturing quality improvement, including prevention of glass breakage and reduction of material loss.

TECHNICAL SKILLS

Programming: Python (scikit-image, NumPy, SciPy, Pandas), OpenCV, SQL, PySpark, Tesseract

Machine Learning: PyTorch, scikit-learn, Hugging Face, LangChain, SHAP, LIME

Data / MLOps / Systems: MLflow, Apache Iceberg, MinIO, Docker, Kubernetes

Operating Systems: Linux, Windows, macOS, ROS

Languages: English (proficient), German (B1)

EDUCATION

Ulm University | Ulm, Germany

12/2018 – 09/2026

Ph.D. in Computer Science | Artificial Intelligence | grade: magna cum laude

Thesis: Learning-based Multimodal Action Recognition

Technical University Kaiserslautern | Kaiserslautern, Germany

10/2015 – 10/2018

M.Sc. in Computer Science | Artificial Intelligence | grade: 1.9

Thesis: Learning 3D Shapes as Multi-Layered Height Maps Using 2D CNN

Visvesvaraya Technological University | Belagavi, India

08/2007 – 06/2011

Bachelor of Engineering in Computer Science | grade: 8.46/10

PUBLICATIONS

1. Temporal Context Enhanced Referring Video Object Segmentation

X. Hu, B. Hampiholi, H. Neumann, J. Lang | WACV 2024

2. Convolutional Transformer Fusion Blocks for Multi-Modal Gesture Recognition

B. Hampiholi, C. Jarvers, W. Mader, H. Neumann | IEEE Access 2023

3. Depthwise Separable Temporal Convolutional Network for Action Segmentation

B. Hampiholi, C. Jarvers, W. Mader, H. Neumann | 3D Vision 2020

4. Learning 3D Shapes as Multi-layered Height Maps using 2D Convolutional Neural Networks

K. Sarkar, B. Hampiholi, K. Varanasi, D. Stricker | ECCV 2018

AWARDS

- Star of the Month — root cause fixes and customer satisfaction (TCS).
- On-the-Spot Award — quick resolution of a critical problem (TCS).
- Best Idea — mobile app development concept (EESTEC Kaiserslautern).